SUBHRANIL GHOSH

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SUMMARY

Java Backend Developer with 4 years of experience building enterprise REST APIs with Java 17, Spring Boot 3 and Spring Batch. Specialized in issuer-side card tokenization (Mastercard MDES, Visa VTS) and secure PIN management with HSM on NCR's card management platform. Maintains 90%+ unit test coverage.

SKILLS

Domain Expertise : Card Tokenization (MDES, VTS), Token Lifecycle Management, Configurable Rule Engine, PIN Management, HSM Integration, Maker-Checker

Languages : Java 8+ (Java 17), SQL

Frameworks : Spring Boot, Spring Batch, Spring Data JPA, Spring Security, Hibernate

Architecture : Microservices, RESTful APIs, Object-Oriented Programming, Swagger/OpenAPI, JSON, XML

Databases : Oracle 19c, PostgreSQL, MySQL, MongoDB

Messaging and Caching : Apache Kafka, Redis

Testing and Quality : JUnit 5, Mockito, Postman, SonarQube, Coverity Scan, Mend (WhiteSource)

Tools and Cloud : Git (GitHub, GitLab), Maven, JIRA, IntelliJ IDEA, Eclipse, VS Code, SQL Developer, MySQL Workbench, pgAdmin, GCP, GitHub Copilot

Methodologies : Agile/Scrum, SDLC (requirements, design, coding, testing, deployment)

WORK EXPERIENCE

Monocept Consulting Pvt. Ltd. | Senior Software Developer Apr 2025 – Present

Promoted from Software Engineer (Nov 2022 – Mar 2025)

- Analyzed client requirements and prepared Low-Level Designs (LLD) to translate business needs into technical specifications.
- Documented APIs with Swagger, giving QA and developers an interactive UI and making API testing roughly 2x faster.
- Designed and developed RESTful APIs using Java 17 and Spring Boot 3.2.2 across multiple services in an Agile environment, delivering sprint commitments on schedule.
- Wrote JUnit 5 and Mockito unit tests, sustaining over 90% code coverage; executed priority (P0) test cases with Postman.
- Monitored code quality with SonarQube, Coverity Scan and Mend (WhiteSource), keeping bugs and code smells at zero in the quality gate.
- Managed source control using Git on GitHub and actively participated in peer code reviews.
- Collaborated closely with QA testers to triage and resolve issues quickly.

Monocept Consulting Pvt. Ltd. | Trainee Software Developer Jul 2022 – Nov 2022

- Completed structured training in Java and microservices development before joining a client project team.

PROJECTS

ACM – Card Management System at NCR Corporation (ongoing client project)

Tech: Java 17, Spring Boot, Spring Batch, Spring Data JPA, REST APIs, Oracle/PostgreSQL, HSM integration, MDES/VTS tokenization

- Developed issuer-side card **tokenization** integrations with Mastercard MDES and Visa VTS.
- Designed and built a configurable **rule engine** that lets the issuer approve, decline, or require additional authentication for token requests.
- Managed the full **token lifecycle**: activate, suspend, resume and terminate.
- Built **PIN management** services that let cardholders view their PIN and raise re-PIN requests.
- Integrated with a Hardware Security Module (HSM) that stores PINs encrypted with two layers of encryption, keeping them highly secure.
- Implemented a generic **Maker-Checker** framework across 5+ services: a maker creates, updates or deletes (CRUD) records, and the change is persisted to the database only after a checker approves it, strengthening validation and security.
- Built configurable report generation with **Spring Batch** for issued cards, expired cards and status changes. Users define templates (header, body, footer fields) and receive reports in CSV or TXT format.
- Contributed to an enterprise-level card management system built on Spring Boot microservices.

EDUCATION

Maulana Abul Kalam Azad University 2020 – 2022

- M.Tech, Information Technology with CGPA: 9.32/10

Adamas Institute of Technology 2015 – 2019

- B.Tech, Computer Science and Engineering with CGPA: 7.63/10